

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 12

Information Theory and the Cost of Description

Assignment

Self-Study Curriculum

Write a one-page analysis of a modeling problem from your own work, structured entirely in information-theoretic language:

1. What is the source (the system generating the data)?
2. What is the entropy of the source (at least approximately)?
3. What model did you use, and what was its description length?
4. What was the KL divergence (or cross-entropy loss, which you can look up from your results)?
5. Could a shorter model (simpler description) have achieved the same or better compression?
6. Was the model capturing the *statistical* regularity or the *generative mechanism* of the system?
How would you know the difference?